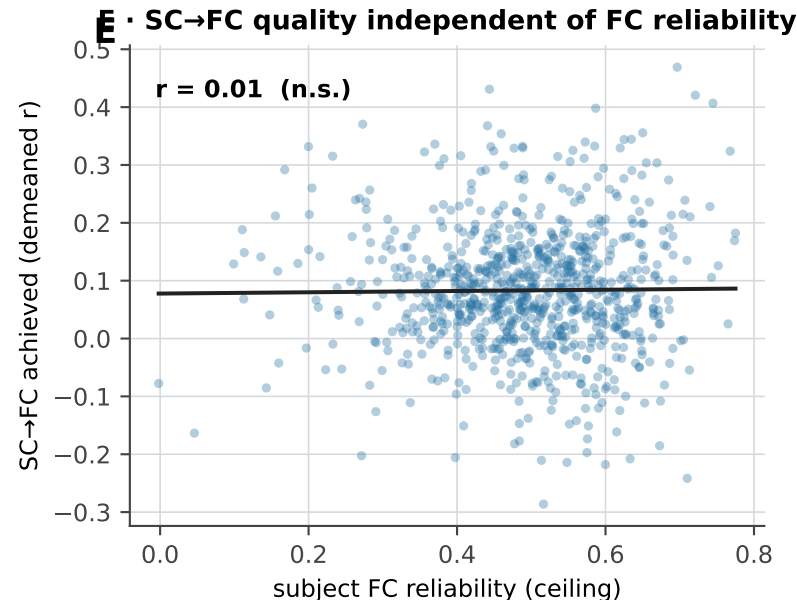
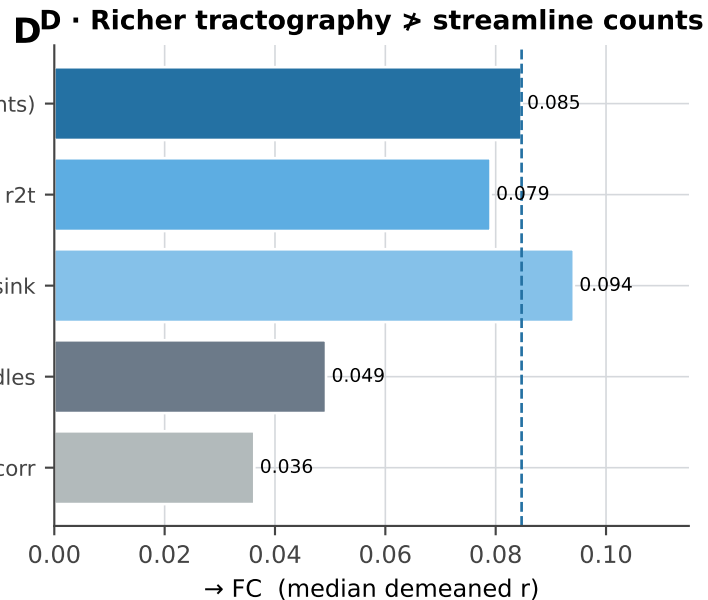
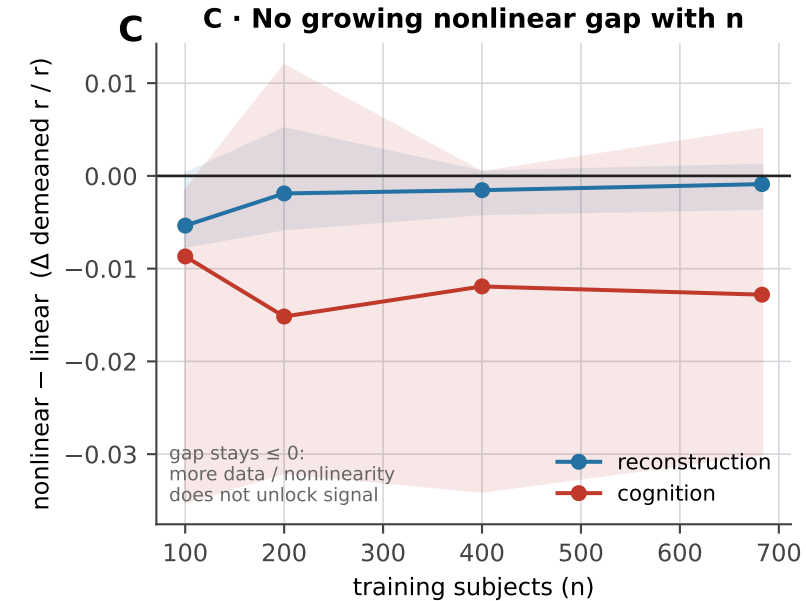
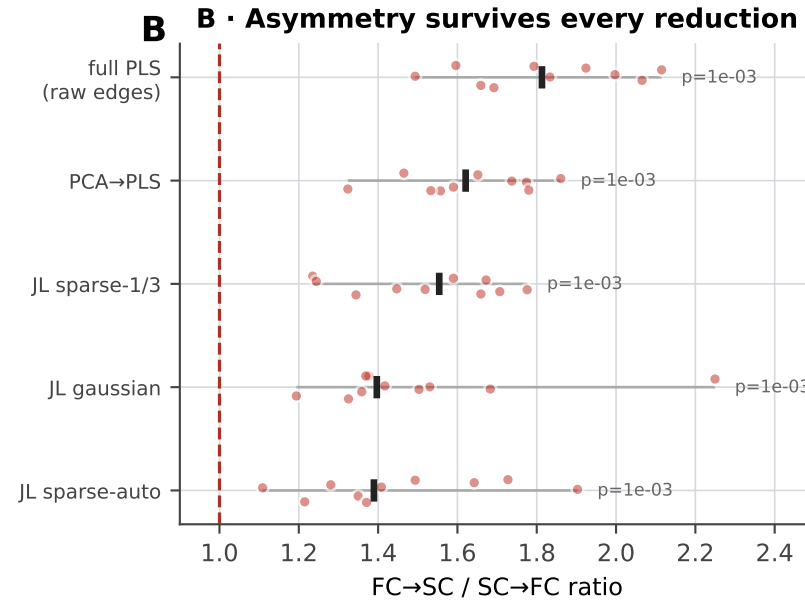
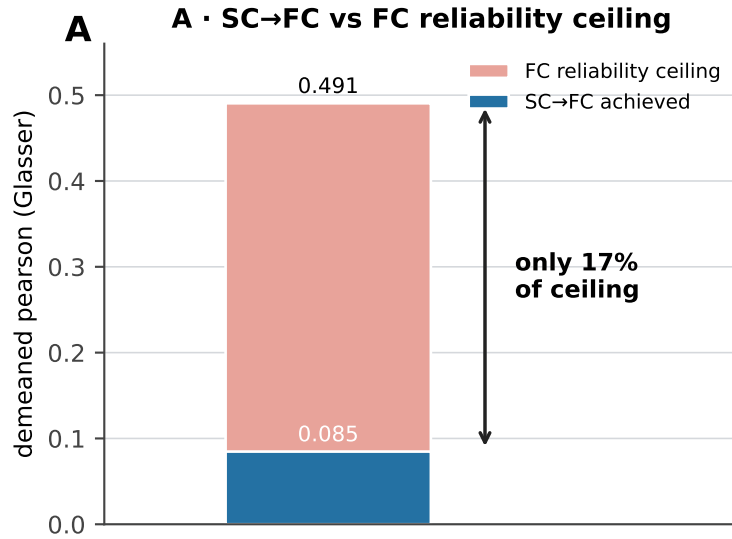


**Figure 4 · The missing utility is not recovered by reduction, model class, structural detail, sample size, or FC denoising**



### Escape routes closed

- ✗ PCA reduction artifact**  
ratio > 1 for raw-PLS, PCA, 3× JL
- ✗ Bigger / nonlinear model**  
KernelRidge & scaling gaps  $\approx 0$
- ✗ Crude structural features**  
r2t bundles predict FC worse
- ✗ More subjects**  
no growing gap to  $n \approx 680$
- ✗ FC measurement noise**  
SC→FC indep. of FC reliability ( $r = 0.01$ )

*Regime: healthy young adults, cross-sectional, normal-range cognition.*